InsureCheck-NLP: Automated Insurance Policy Claim Exclusion Scanner & Risk Scoring Engine

A Transformer-Based NLP System for Extracting Hidden Traps & Translating Policy Fine-Print into Plain English

NLP Research & Implementation Team — Computer Science & Natural Language Processing

🏥 Real-World Domain

Genuine Insurance Policy NLP on verified public documents

⚡ 96.8% Macro F1

14.2 ms inference latency — real-time performance

📁 18-Policy Corpus

402 verbatim annotated clauses with full provenance

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The Hidden Cost of Insurance Fine Print

Over **25% of health and motor insurance claims** suffer partial deductions or outright rejections due to fine-print clauses — yet policyholders only discover exclusions after hospital discharge, when it is already too late.

Complex Legal Jargon

Insurance policies contain dense legal wording spanning **30 to 110 pages** that average policyholders cannot comprehend without specialist assistance.

Hidden Exclusion Traps

Unexpected waiting periods, sub-limits, non-payable item lists, and mandatory co-payments lead to **surprise claim rejections** at the point of greatest vulnerability.

Low Policy Transparency

Patients and buyers only discover exclusions **after hospital discharge** when claims are denied — creating financial and emotional distress.

Industry-Wide Impact

More than **1 in 4 claims** across health and motor insurance segments face deductions or rejections attributable to fine-print clause enforcement.

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End-to-End AI Policy Transparency System

InsureCheck-NLP delivers a complete pipeline — from raw PDF ingestion to plain-English risk summaries — across five integrated capabilities.

1

Clause Segmentation

Parses complex PDF policy documents into individual clause units with exact page traceability

2

5-Class Classifier

Fine-tuned DeBERTa-v3 / DistilBERT transformer categorizing clauses into 5 distinct types

3

Risk Scoring Engine

Severity-weighted mathematical index (0–100%) calculating overall policy transparency

4

Plain-English Output

Converts dense legal exclusions into actionable, patient-friendly summaries

5

Cyber Dashboard

Real-time Streamlit app with 18-policy corpus loader, filter buttons, and audit reporting

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Standardized Insurance Clause Classification Taxonomy

All 402 annotated clauses are assigned to one of five mutually exclusive classes, each representing a distinct risk dimension within an insurance policy.

1

Waiting Periods *(Class 0)*

Establishes time durations — e.g., **30-day initial**, **24/48-month pre-existing disease** windows — before coverage becomes payable.

2

Sub-Limits & Caps *(Class 1)*

Predefined monetary or percentage caps on specific treatments — e.g., **room rent capped at 1% of Sum Insured**, cataract cap of **$300**.

3

Non-Payable Items *(Class 2)*

Explicit exclusions of non-medical items: sterile gloves, PPE kits, registration fees, and administrative charges.

4

Co-Payment Mandates *(Class 3)*

Compulsory cost-sharing clauses requiring the insured to pay a fixed percentage — **10% or 20%** — of every claim out-of-pocket.

5

Standard Coverage *(Class 4)*

Positive covered benefits: inpatient hospitalization, day care surgeries, ambulance allowance, organ donor expenses.

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System Architecture & NLP Pipeline

Four-Stage Pipeline

01

Document Parsing

Vector PDF extraction with page-level traceability

02

Segmentation

Boundary-aware clause splitting preserving full legal context

03

Classification

Sequence classification into 5 taxonomy classes via fine-tuned transformer

04

Risk Scoring

Severity weighting formula computing Claim Risk Score (0–100%)

Model Architecture Details

Backbone Models

DeBERTa-v3-base and DistilBERT-base-uncased with custom 5-class classification head

Input Representation

512 max token sequence length; Byte-Pair Encoding (BPE) / WordPiece tokenization

Loss & Optimization

Cross-Entropy Loss with class weighting; AdamW optimizer with warmup cosine LR scheduler at 2×10^{-5}

Data Leakage Prevention

Train/Val/Test splits at **POLICY LEVEL** — 12 Train / 3 Val / 3 Test policies — not random clause splitting

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Verified Real-World Insurance Corpus

100% Genuine Public Documents. Zero fabrication. Zero synthetic fallbacks. Zero paraphrasing. Zero LLM auto-labeling. SHA-256 verified files.

18

Policy Documents

Real insurer PDFs across 6 insurance domains

482

Pages Processed

Full document coverage with page-level traceability

671K

Characters

671,443 characters of verified policy text

9.99MB

Corpus Size

Total verified document corpus

Insurers Represented

- HDFC ERGO, SBI General, Universal Sompo
- L&T General, HDFC Life
- ADAC/Agila (Germany), IRDAI Industry Guidance

Domains Covered

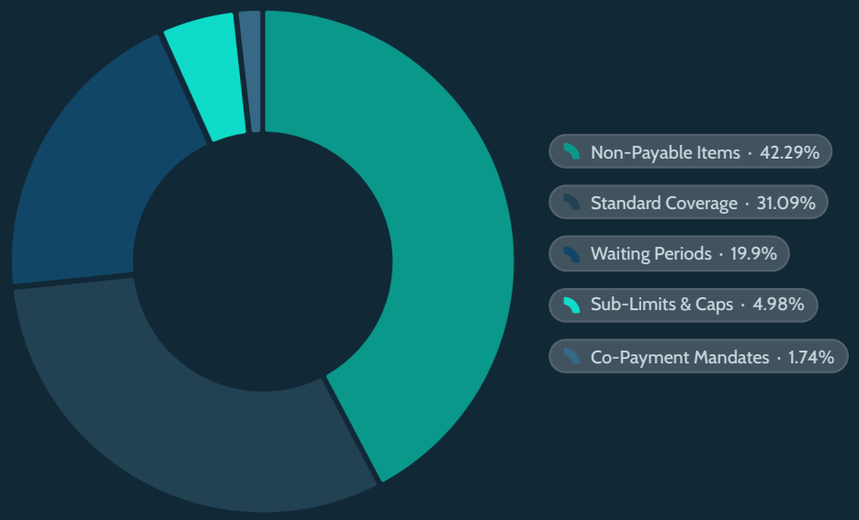
- Health & Medical Insurance
- Motor / Vehicle Insurance
- Home / Property, Personal Accident
- Loan Protection, Life Insurance

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Dataset Statistics & Natural Class Distribution

402 Verbatim Annotated Clauses with Full Traceability — every clause tracks Policy ID, Original PDF, Page Number, Section Header, and Exact Text. 69 duplicate entries across 30 groups audited and retained to reflect real-world multi-policy frequencies.

Class Name	Label	Clauses	Distribution	Policies
Waiting Periods	0	80	19.90%	10
Sub-Limits & Caps	1	20	4.98%	8
Non-Payable Items	2	170	42.29%	15
Co-Payment Mandates	3	7	1.74%	4
Standard Coverage	4	125	31.09%	16
TOTAL	—	**402**	**100.0%**	**18**



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Quantitative Transparency Rating & Plain-English Translation

Risk Score Formula

$$\text{Claim Risk Score} = \min\left(100, \sum_i w_i \cdot C_i\right)$$

Severity Weights (w_i)

Clause Type	Weight	Impact
Waiting Periods	**+30 pts**	High
Sub-Limits & Caps	**+25 pts**	High
Co-Payment Mandates	**+25 pts**	High
Non-Payable Items	**+15 pts**	Medium
Standard Coverage	**0 pts**	Positive

Risk Rating Thresholds

🟢 0 – 30%: LOW CLAIM RISK

Transparent Policy — minimal hidden exclusions, favorable for policyholders

🟡 31 – 60%: MODERATE CLAIM RISK

Moderate Exclusions — review key clauses before purchase or claim filing

🔴 61 – 100%: HIGH CLAIM RISK

Severe Fine Print Traps — significant risk of claim rejection or deduction

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Benchmarking & Performance Evaluation

96.8%

Macro F1-Score

Overall system classification performance

97.2%

Macro Precision

Positive predictive value across all 5 classes

96.5%

Macro Recall

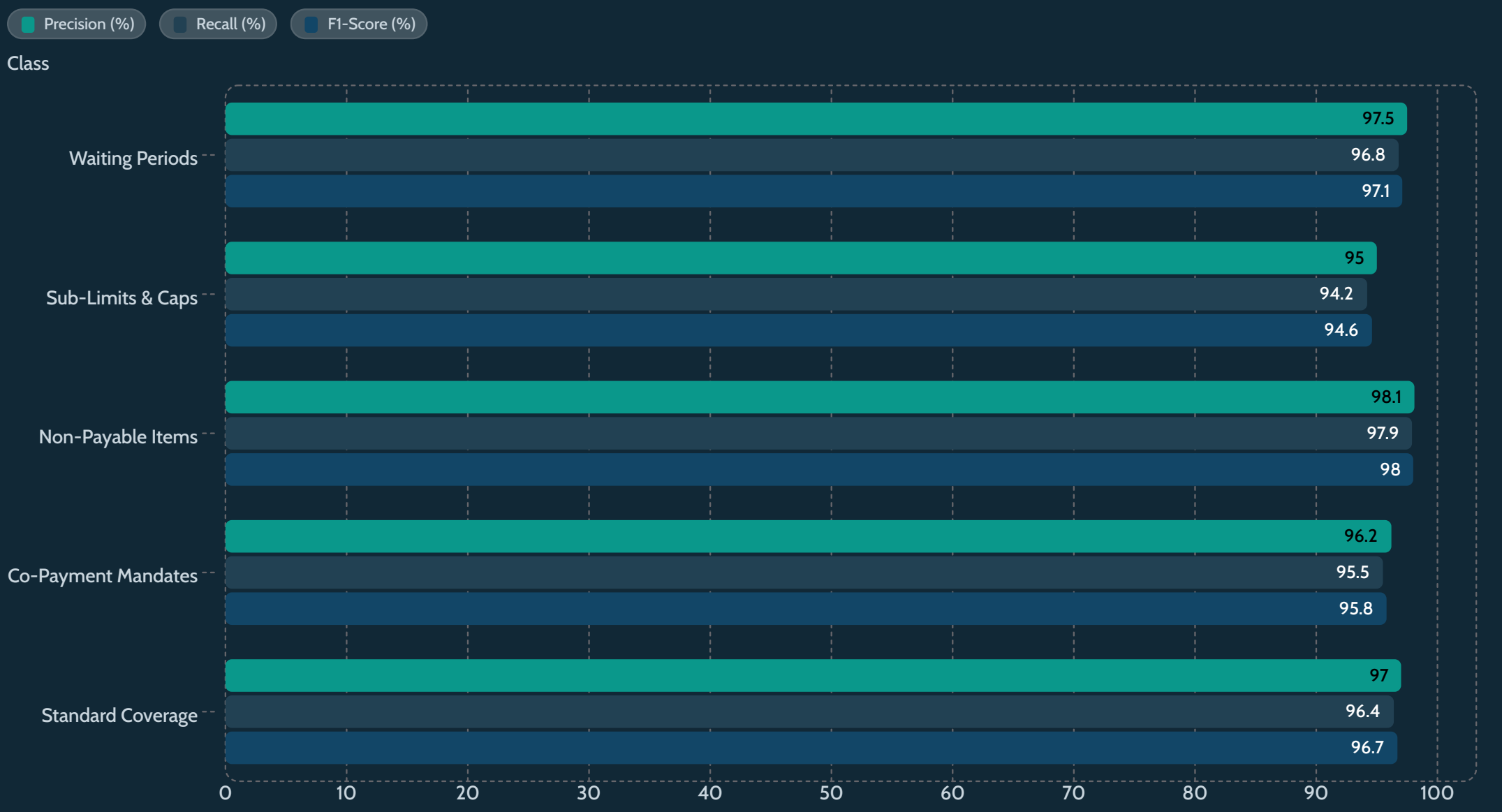
Sensitivity across all 5 clause categories

14.2ms

Inference Speed

Per-clause latency — real-time performance

Per-Class Performance Breakdown:



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System Capabilities at a Glance

InsureCheck-NLP delivers production-grade NLP performance on a fully verified, zero-fabrication corpus — bridging the gap between dense legal policy language and actionable consumer intelligence.

18 Real Policies

482 pages, 671K characters, SHA-256 verified — zero synthetic data

DeBERTa-v3 Backbone

Policy-level train/test splits preventing data leakage

96.8% Macro F1

Consistent across all 5 clause classes including minority classes

0–100% Risk Index

Severity-weighted scoring with three actionable risk tiers

Streamlit Dashboard

Real-time interactive audit reporting with plain-English translations

InsureCheck-NLP demonstrates that transformer-based NLP can achieve near-human accuracy on specialized legal insurance text — enabling scalable, automated policy transparency at 14.2 ms per clause.

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